

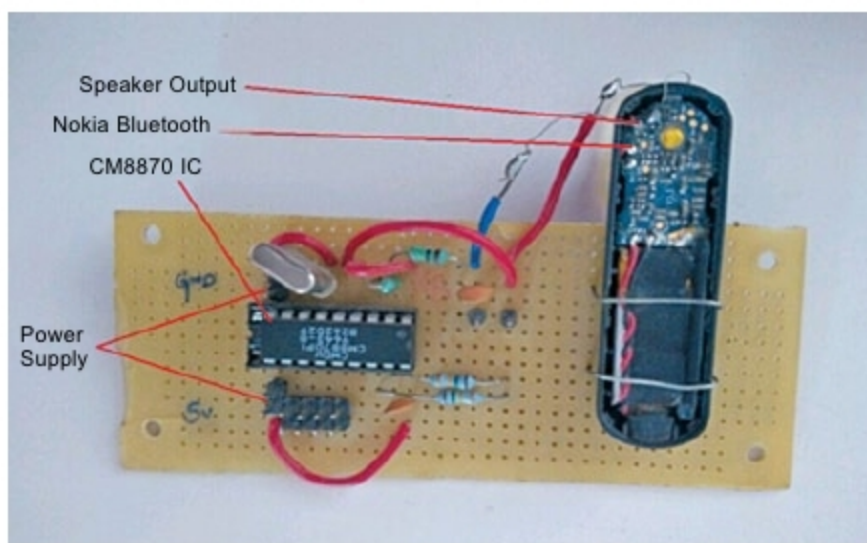


Fig. 3: Bluetooth headset connections with MT8870

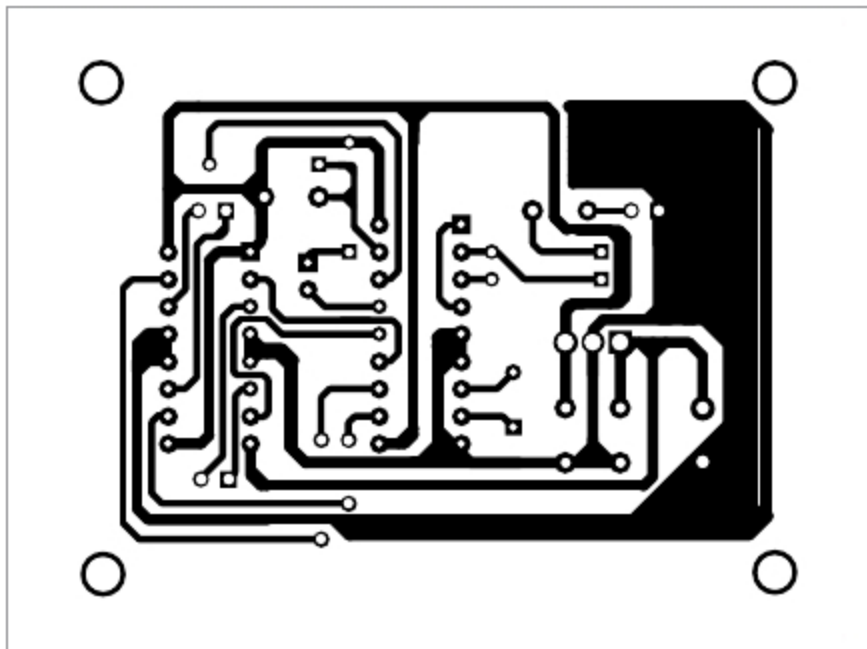


Fig. 4: Actual-size PCB layout of robot controller

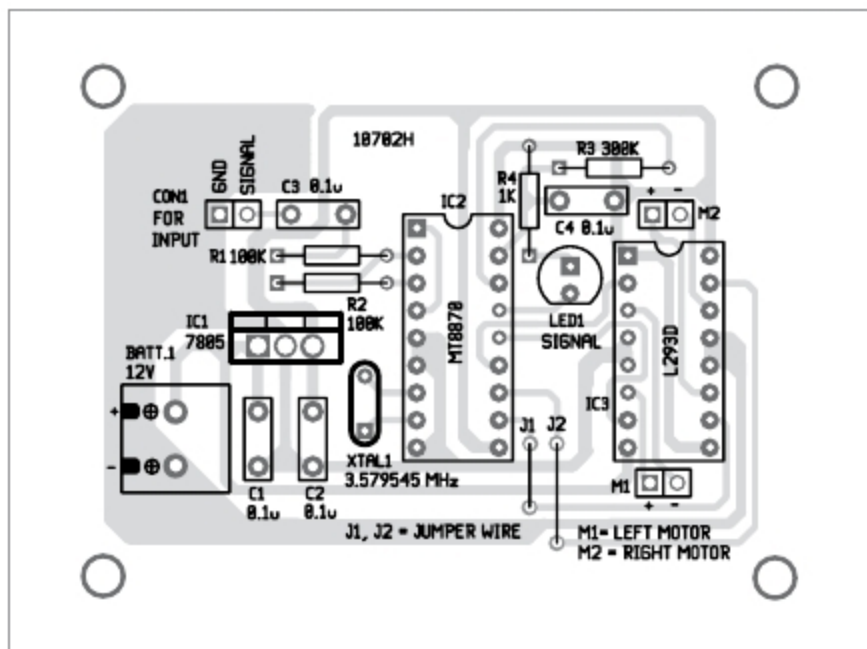


Fig. 5: Components layout for the PCB

## Construction and testing

An actual-size PCB layout of the robot is shown in Fig. 4 and its components layout in Fig. 5. The robot car is built using two geared motors for rear wheels and a castor wheel on the front. Connect the left and right

motors (M1 and M2) to output pins (out1 through out4) of L293D motor driving circuit. Outputs Q1 through Q4 of DTMF IC2 are connected to L293D inputs.

Power supply is provided through a 12V lead-acid battery. For MT8870 and L293D, power supply is provided via 7805 voltage regulator. Bluetooth headset has an inbuilt battery that can be charged using an ordinary phone charger. Charge Bluetooth headset and then pair it with the phone's Bluetooth.

There are many apps available for DTMF tone generation, such as DTMF by Wolphi LLC. For customisation, or to build your own app of DTMF application, MIT App Inventor can be used. Tonedef app for Android was used during testing at EFY Lab. With this app, you need two cellphones: one (say, A) with DTMF app installed and in user's hand, and the second cellphone (say, B) with auto answering mode enabled and wirelessly connected to Nokia-BH-108 Bluetooth headset.

The robot is controlled from the first cellphone after connection is established with the second

cellphone. Advantage here is that you do not have to keep the second phone on the robot along with Nokia BH-108 Bluetooth headset.

Keep both cellphones with you. Dial the number of phone B from phone A. Once calling is established

## PARTS LIST

### Semiconductors:

|      |                                |
|------|--------------------------------|
| IC1  | - 7805, 5V voltage regulator   |
| IC2  | - MT8870 DTMF receiver/decoder |
| IC3  | - L293D motor driver           |
| LED1 | - 5mm LED                      |

### Resistors (all 1/4-watt, $\pm 5\%$ carbon):

|       |                |
|-------|----------------|
| R1-R2 | - 100-kilo-ohm |
| R3    | - 300-kilo-ohm |
| R4    | - 1-kilo-ohm   |

### Capacitors:

|       |                            |
|-------|----------------------------|
| C1-C4 | - 0.1 $\mu$ F ceramic disk |
|-------|----------------------------|

### Miscellaneous:

|        |                                                    |
|--------|----------------------------------------------------|
| BATT.1 | - 12V battery                                      |
| M1-M2  | - 12V DC geared motor, 50RPM                       |
| XTAL1  | - 3.579545MHz crystal oscillator                   |
| CON1   | - 2-pin connector for input from Bluetooth headset |
|        | - Nokia BH-108 or equivalent Bluetooth headset     |
|        | - Cellphone                                        |
|        | - 18-pin and 16-pin IC socket                      |
|        | - Heat-sink for 7805                               |

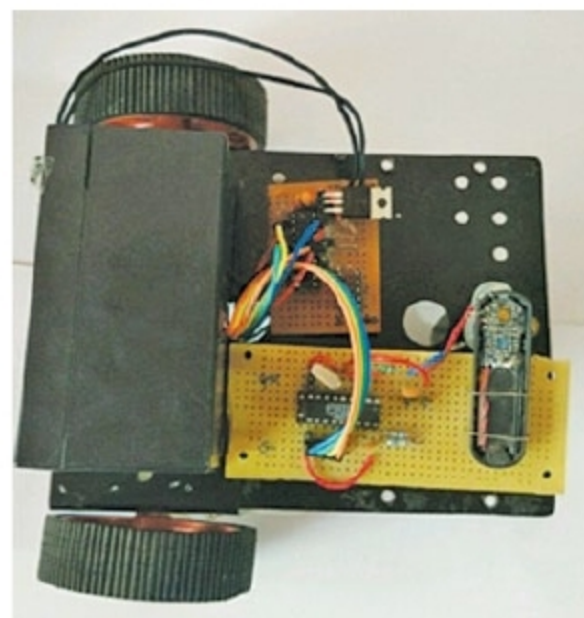


Fig. 6: Author's prototype of the robot

between the two phones, open DTMF app on phone A. You will see a virtual dialpad. Touch the appropriate number as per the table to control the robot in forward, backward, right or left directions. DTMF signal from phone A is transmitted and received by DTMF decoder MT8870 through BH-108 Bluetooth headset mounted on the robot's chassis. The author's prototype is shown in Fig. 6.

**Note.** You can use any suitable Bluetooth headset for this project. **EFY**



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